Microwave Filter for CGM Devices in Automatic Insulin Delivery System to Prevent Pancreas Thermogenesis Noise Signal

Supplementary

1. Research Mapping

Table 1:
Mapping of Existing Literature vs Remarks

Ref	Remarks	Identified Research Gap	Proposed Work
CSRR antenna (1.02–2.23 GHz) provides high penetration (20–25 mm) and strong correlation ($R^2 > 0.98$) from forearm [36]	Excessive penetration excites non-target tissues, causing interference	Lack of selectivity for deep-organ glucose sensing	AHRF isolates glucose-sensitive resonant band and suppresses broadband tissue radiation
MTM sensors (2.185 GHz) achieve shallow penetration (~5 mm) [37]	Limited depth, invasive blood-sample-based sensing	No capability for pancreas-level sensing (30–45 mm depth)	High-Q AHRF enables controlled penetration with improved signal confinement
RDR sensor (2.47 GHz) achieves good accuracy but >35–45 dB signal loss [38]	Severe attenuation reduces sensitivity below 0.05 MHz/(mg/dL)	Insufficient frequency sensitivity for clinical tolerance	AHRF increases effective Q (>200) to boost sensitivity above 0.5 MHz/(mg/dL)
Fingertip NIBGM sensors suffer from weak dielectric contrast ($\Delta\varepsilon_r \approx 0.5$) and high MARD (>20%) [39]	Peripheral sensing distorted by tissue attenuation and SAR limits	Poor correlation with real-time insulin dynamics	Pancreatic sensing using SAIH antenna integrated with AHRF
Saliva-based implanted antenna shows weak correlation ($r < 0.7$) and ultra-low glucose levels [40]	Biochemical variability and lack of stability	Non-clinically reliable sensing medium	Direct pancreatic dielectric sensing avoids biochemical dilution
HSIS/HI antennas demonstrate high pancreatic accuracy (>85%) [19], [20]	Noise sources (thermal, thermogenesis, motion) not addressed	Signal corruption at deep tissue level	Noise-aware AHRF-integrated antenna system

Table 2:
Noise to solution mapping

Identified Problem	Noise Source	Quantitative Impact	Proposed Work
Temperature variation in pancreas	Thermogenesis noise	$\sim 1\times10^{-17}$ W (100 kHz BW)	Resonant-band isolation using AHRF
Random electronic fluctuations	Thermal noise	~−124 dBm @ 310 K	Bandwidth narrowing via coupling gap optimization
Organ displacement (1–5 mm)	Motion artefact noise	±10–30% signal variation	U-shaped AHRF geometry for motion robustness

Existing antennas as sensors for CGM devices mostly focus on peripheral glucose sensing. When used on deep organs like the pancreas, they produce too much attenuation, insufficient dielectric contrast, and noise. HSIS and HI antennas as sensors validate capable pancreatic accuracy the absence of high-Q filtering and noise-aware design limits their clinical robustness. The proposed method bridges the gap by coupling an AHRF with a pancreas optimised SAIH antenna to reduce thermogenesis noise, thermal noise (by reducing the bandwidth), and motion artefacts via structural symmetry. The proposed method has a frequency sensitivity of more than 0.5 MHz/(mg/dL) and can detect glucose levels that are not clinically relevant within a ±5 mg/dL range.

2. Contribution

Table 3:
Contribution mapping

Ref	Contribution
[38]/2025	1. Implement two-port rectangular dielectric resonator (RDR) microwave sensor for noninvasive glucose detection. 2. Developed a complete standalone CGM system using VCO and power detector, eliminating the need for VNA. 3. Introduced Cole–Cole–based finger tissue modeling for realistic simulation. 4. Achieved high resolution and sensitivity with strong experimental correlation to invasive glucometers (>92% accuracy).
[37]/2025	1. Designed blood samples using metamaterial microwave sensors. 2. Utilized phase variation of transmission coefficient S_{21} for glucose prediction. 3. Gender specific mathematical models to account for physiological dielectric differences. 4. Evaluate performance using MARD and Parkes Error Grid analysis, confirming feasibility for noninvasive monitoring.
[36]/2025	1. Developed CSRR-loaded antenna sensor with improved compactness and sensitivity. 2. Multifrequency narrowband operation for enhanced robustness against environmental interference. 3. Machine learning regression models, achieving glucose prediction accuracy above 98% (R^2).
[27]/2022	1. Developed a wearable flexible antenna sensor integrated into a thumb spica splint for continuous monitoring. 2. Introduced dielectric radiation based glucose sensing, reducing skin and muscle interference. 3. 96.8% glucose prediction accuracy, validated against invasive glucometer measurements.
Proposed	***1. To propose, Asymmetrical Hairpin Resonator Filter (AHRF), reduces thermogenesis noise through isolating the resonant band, eliminates tissue-induced broadband fluctuations of proposed SAIH antenna and the acquired signal.*** ***2. To reduce thermal noise, narrow bandwidth in AHRF design is included and obtain signal through radiation coupling gap between adjacent hairpin.*** ***3. To eliminate motion artefacts in antenna acquired signal, U-shaped is proposed in the AHRF filter design.*** ***4. To evaluate the efficiency of AHRF filters, compared with Coupled Line Bandpass Filter (CLBPF)***

3. Data Collection

Proposed SAIH antenna sensor with AHRF filter and their signals provides high accuracy in CGM device. This is based on the study from Jan 2025 to Dec 2025 500 diabetes patients at Jebi Medical Center in Coimbatore, Tamil Nadu, India. The Ethics Committee approval on March 14, 2022 (EC/2852/2022). Nearly, 400 people had to fast for 8 to 10 hours and drink water before acquire of signals. During fasting measurement, they drank glucose dissolved in water and recorded using SAIH and AHRF again an hour later. Before the signal was acquired, the last hundred people ate whatever they want. Table 4 shows the percentage distribution of energy bands before food intake and Table 5 shows the percentage distribution of energy bands after food intake.

Different age groups and glucose level testing, with a reliable reference link included:

https://github.com/Jefferson-2012/Jebasingh.git

Table 4:

Percentage distribution of energy bands before food intake

Patient _ID	Age	Gender	Lab Value (mg/dL)	Subband_1	Subband_2	Subband_3	Subband_4	Subband_5	Subband_6	Subband_7	Subband_8	Subband_9	Subband_10	Subband_11
P 001	43	Female	100	20.273	4.351	4.703	3.936	3.913	4.146	5.886	5.42	5.383	5.474	4.889
P 002	81	Male	114	23.182	6.181	5.258	4.319	5.658	4.99	5.479	4.8	4.579	5.383	4.984
P 003	29	Male	96	18.508	3.89	5.033	4.812	3.223	5.549	4.174	3.946	4.574	5.295	5.234
P 004	21	Male	118	23.231	6.781	4.948	5.339	5.008	4.084	4.933	4.909	5.001	4.859	4.859
P 005	23	Female	127	25.331	5.926	2.261	5.319	6.911	4.491	4.905	5.095	5.244	5.222	4.951
P 006	23	Male	135	27.314	6.774	2.207	3.892	5.442	4.809	5.719	4.7	4.837	4.459	4.875
P 007	47	Female	144	29.077	8.328	4.731	5.839	5.208	5.911	5.339	4.635	4.932	4.927	5.041
P 008	84	Male	119	23.95	5.795	3.721	5.501	4.773	5.429	5.028	5.01	5.369	4.874	4.761
P 009	68	Female	122	24.396	6.271	4.293	5.984	5.813	4.852	4.614	4.735	5.267	5.01	4.845
P 010	43	Female	98	19.79	3.557	4.535	5.368	4.048	4.374	4.615	5.729	4.748	5.189	4.782
P 011	76	Female	100	20.178	4.242	4.989	4.536	4.514	5.888	5.75	5.42	5.648	4.969	5.178
P 012	76	Female	103	20.545	4.154	3.999	5.704	5.166	5.093	5.092	4.722	4.941	4.917	4.822
P 013	50	Male	102	19.988	4.427	5.342	3.93	5.857	4.187	3.978	5.081	4.64	4.695	4.837
P 014	70	Female	101	20.128	4.441	5.155	4.939	6.417	3.97	5.573	5.292	5.365	4.326	4.781
P 015	39	Male	105	21.349	5.178	6.038	3.193	4.479	5.534	4.009	4.99	4.988	4.787	5.032
P 016	38	Male	97	19.509	4.803	6.51	5.71	4.089	4.199	4.476	5.892	5.618	5.049	5.027
P 017	21	Male	110	22.409	5.586	5.308	5.531	5.325	5.093	4.841	6.015	4.464	4.732	4.883
P 018	79	Male	107	21.522	4.148	3.343	4.018	5.626	5.78	4.829	4.969	4.998	5.33	5.266
P 019	79	Female	116	22.738	5.8	4.269	4.971	6.115	4.621	4.529	5.394	5.317	4.62	5.249
P 020	41	Female	87	16.502	3.491	4.477	4.414	5.248	3.886	4.356	4.991	5.588	5.438	4.831
P 021	72	Male	119	23.853	5.876	4.787	7.153	5.799	4.011	4.562	5.406	5.067	5.33	4.948
P 022	26	Female	107	21.324	4.757	4.722	7.298	5.73	4.63	4.692	4.36	4.601	4.781	5.005
P 023	20	Male	97	19.052	5.167	6.494	5.517	4.843	5.798	6.728	4.499	5.483	5.21	4.892
P 024	48	Male	91	17.594	3.035	3.271	6.501	5.207	4.355	4.183	5.526	5.38	5.219	5.094
P 025	57	Female	87	17.138	3.73	6.201	4.876	5.009	4.124	5.658	5.482	5.075	4.788	4.909
P 026	53	Male	115	22.931	6.175	5.485	5.904	5.423	3.414	4.352	4.749	4.993	4.946	5.132

Patient _ID	Age	Gender	Lab Value (mg/dL)	Subband_1	Subband_2	Subband_3	Subband_4	Subband_5	Subband_6	Subband_7	Subband_8	Subband_9	Subband_10	Subband_11
P 027	41	Male	121	24.437	7.029	5.548	4.712	5.273	5.604	4.211	5.164	5.8	5.274	4.886
P 028	23	Female	95	19.067	4.725	5.426	4.029	5.018	4.642	4.695	4.797	4.418	4.81	4.99
P 029	83	Male	112	22.973	5.604	5.19	6.327	5.789	4.791	4.67	4.702	4.685	4.53	5.239
P 030	68	Male	89	17.817	3.135	4.442	6.139	4.575	4.867	5.303	4.012	4.686	4.867	4.848
P 031	79	Female	88	17.847	3.949	4.985	5.576	3.698	4.961	4.792	4.932	4.667	4.768	5.12
P 032	74	Female	97	19.245	4.583	5.708	5.265	5.947	4.855	4.969	5.242	5.401	4.684	4.97
P 033	83	Female	114	23.182	6.402	4.918	5.143	5.497	4.175	5.602	5.184	4.792	5.387	5.257
P 034	25	Male	88	17.705	3.632	4.864	4.397	4.208	5.251	5.409	5.088	4.995	4.876	5.172
P 035	43	Male	112	22.158	4.707	4.191	5.187	5.082	4.712	4.626	4.822	5.251	5.029	4.927
P 036	62	Male	104	20.715	5.066	5.573	3.819	4.687	4.533	5.696	5.611	4.73	4.928	4.998
P 037	61	Male	115	23.174	5.451	3.801	3.709	4.079	4.729	4.678	5.116	4.474	5.156	5.477
P 038	22	Male	102	19.98	4.678	5.819	5.943	5.239	5.287	4.957	4.62	4.961	4.799	5.207
P 039	43	Male	121	24.19	5.788	4.248	5.374	4.241	5.042	4.635	4.839	4.964	5.368	4.868
P 040	84	Female	119	23.941	5.929	4.476	5.117	5.368	4.788	6.102	5.326	5.509	4.605	5.256
P 041	36	Female	100	20.012	4.509	5.023	4.342	5.149	4.928	5.003	4.673	4.777	5.144	5.337
P 042	37	Female	123	25.08	6.797	5.513	4.049	5.599	4.362	5.021	4.444	4.857	4.848	4.783
P 043	29	Male	113	22.876	5.591	4.843	5.747	3.741	5.328	4.495	4.498	4.758	4.878	4.833
P 044	64	Female	105	20.997	4.364	4.409	4.738	3.865	4.529	5.02	5.488	4.662	4.892	5.151
P 045	22	Female	108	21.677	5.603	4.86	3.876	5.287	6.434	4.812	5.159	4.649	5.125	5.042
P 046	31	Male	100	20.105	4.465	5.016	5.457	4.617	5.016	4.466	4.95	4.895	4.696	4.85
P 047	81	Male	95	19.144	4.845	5.754	4.419	4.05	5.001	4.117	5.384	4.597	4.826	5.257
P 048	55	Female	110	21.35	5.612	5.637	5.945	4.66	4.438	5.034	5.033	5.205	4.853	4.918
P 049	54	Female	103	20.493	3.956	4.083	6.134	5.412	4.771	5.749	4.877	5.161	4.694	5.042
P 050	28	Female	112	21.871	4.654	2.955	4.985	4.639	4.936	5.709	4.17	4.425	4.968	5.22

Table 5:

Percentage distribution of energy bands after food intake

Patient _ID	Age	Gender	Lab Value (mg/dL)	Subband_1	Subband_2	Subband_3	Subband_4	Subband_5	Subband_6	Subband_7	Subband_8	Subband_9	Subband_10	Subband_11
P 001	43	Female	248	31.546	4.851	5.153	4.686	4.863	4.996	4.906	6.400	4.533	4.724	4.239
P 002	81	Male	342	33.225	6.681	5.708	5.069	6.608	5.840	4.499	5.780	3.729	4.633	4.334
P 003	29	Male	243	31.437	4.742	5.439	5.286	5.464	6.738	4.770	6.400	4.798	4.219	4.528
P 004	21	Male	230	31.085	4.390	5.483	5.562	4.173	6.399	3.194	4.926	3.724	4.545	4.584
P 005	23	Female	229	31.349	4.654	4.449	6.454	6.116	5.943	4.112	5.702	4.091	4.167	4.172
P 006	23	Male	246	31.622	4.927	5.792	4.680	6.807	5.037	2.998	6.061	3.790	3.945	4.187
P 007	47	Female	263	31.636	4.941	5.605	5.689	7.367	4.820	4.593	6.272	4.515	3.576	4.131
P 008	84	Male	300	32.373	5.678	6.488	3.943	5.429	6.384	3.029	5.970	4.138	4.037	4.382
P 009	68	Female	218	31.197	5.303	6.960	6.460	5.039	5.049	3.496	6.872	4.768	4.299	4.377
P 010	43	Female	322	32.781	6.086	5.758	6.281	6.275	5.943	3.861	6.995	3.614	3.982	4.233
P 011	76	Female	227	31.343	4.648	3.793	4.768	6.576	6.630	3.849	5.949	4.148	4.580	4.616
P 012	76	Female	325	32.995	6.300	4.719	5.721	7.065	5.471	3.549	6.374	4.467	3.870	4.599
P 013	50	Male	171	30.294	3.991	4.927	5.164	6.198	4.736	3.376	5.971	4.738	4.688	4.181
P 014	70	Female	316	33.071	6.376	5.237	7.903	6.749	4.861	3.582	6.386	4.217	4.580	4.298
P 015	39	Male	251	31.952	5.257	5.172	8.048	6.680	5.480	3.712	5.340	3.751	4.031	4.355
P 016	38	Male	287	32.362	5.667	6.944	6.267	5.793	6.648	5.748	5.479	4.633	4.460	4.242
P 017	21	Male	155	30.230	3.535	3.721	7.251	6.157	5.205	3.203	6.506	4.530	4.469	4.444
P 018	79	Male	209	30.925	4.230	6.651	5.626	5.959	4.974	4.678	6.462	4.225	4.038	4.259
P 019	79	Female	343	33.370	6.675	5.935	6.654	6.373	4.264	3.372	5.729	4.143	4.196	4.482
P 020	41	Female	389	34.224	7.529	5.998	5.462	6.223	6.454	3.231	6.144	4.950	4.524	4.236
P 021	72	Male	279	31.920	5.225	5.876	4.779	5.968	5.492	3.715	5.777	3.568	4.060	4.340
P 022	26	Female	316	32.799	6.104	5.640	7.077	6.739	5.641	3.690	5.682	3.835	3.780	4.589
P 023	20	Male	388	33.976	7.281	5.398	6.089	5.958	4.934	3.953	5.889	4.151	4.109	4.209
P 024	48	Male	374	33.969	7.274	2.657	4.642	6.392	5.659	4.739	5.680	3.987	3.709	4.225
P 025	57	Female	166	30.330	3.635	4.892	6.889	5.525	5.717	4.323	4.992	3.836	4.117	4.198
P 026	53	Male	220	31.144	4.449	5.435	6.326	4.648	5.811	3.812	5.912	3.817	4.018	4.470

Patient _ID	Age	Gender	Lab Value (mg/dL)	Subband_1	Subband_2	Subband_3	Subband_4	Subband_5	Subband_6	Subband_7	Subband_8	Subband_9	Subband_10	Subband_11
P 027	41	Male	246	31.778	5.083	6.158	6.015	6.897	5.705	3.989	6.222	4.551	3.934	4.320
P 028	23	Female	369	33.597	6.902	5.368	5.893	6.447	5.025	4.622	6.164	3.942	4.637	4.607
P 029	83	Male	195	30.827	4.132	5.314	5.147	5.158	6.101	4.429	6.068	4.145	4.126	4.522
P 030	68	Male	262	31.902	5.207	4.641	5.937	6.032	5.562	3.646	5.802	4.401	4.279	4.277
P 031	79	Female	352	33.466	6.771	4.743	6.734	6.763	5.702	3.634	5.715	4.417	4.260	4.195
P 032	74	Female	303	32.261	5.566	6.023	4.569	5.637	5.383	4.716	6.591	3.880	4.178	4.348
P 033	83	Female	309	32.646	5.951	4.251	4.459	5.029	5.579	3.698	6.096	3.624	4.406	4.827
P 034	25	Male	253	31.873	5.178	6.269	6.693	6.189	6.137	3.977	5.600	4.111	4.049	4.557
P 035	43	Male	336	32.983	6.288	4.698	6.124	5.191	5.892	3.655	5.819	4.114	4.618	4.218
P 036	62	Male	326	33.124	6.429	4.926	5.867	6.318	5.638	5.122	6.306	4.659	3.855	4.606
P 037	61	Male	255	31.704	5.009	5.473	5.092	6.099	5.778	4.023	5.653	3.927	4.394	4.687
P 038	22	Male	371	33.992	7.297	5.963	4.799	6.549	5.212	4.041	5.424	4.007	4.098	4.133
P 039	43	Male	305	32.786	6.091	5.293	6.497	4.691	6.178	3.515	5.478	3.908	4.128	4.183
P 040	84	Female	246	31.559	4.864	4.859	5.488	4.815	5.379	4.040	6.468	3.812	4.142	4.501
P 041	36	Female	325	32.798	6.103	5.310	4.626	6.237	7.284	3.832	6.139	3.799	4.375	4.392
P 042	37	Female	252	31.660	4.965	5.466	6.207	5.567	5.866	3.486	5.930	4.045	3.946	4.200
P 043	29	Male	272	32.040	5.345	6.204	5.169	5.000	5.851	3.137	6.364	3.747	4.076	4.607
P 044	64	Female	315	32.807	6.112	6.087	6.695	5.610	5.288	4.054	6.013	4.355	4.103	4.268
P 045	22	Female	206	31.151	4.456	4.533	6.884	6.362	5.621	4.769	5.857	4.311	3.944	4.392
P 046	31	Male	255	31.849	5.154	3.405	5.735	5.589	5.786	4.729	5.150	3.575	4.218	4.570
P 047	81	Male	208	31.024	4.329	5.543	5.970	7.291	6.982	4.018	6.146	4.724	3.902	4.284
P 048	55	Female	251	31.509	4.814	6.272	5.778	6.693	5.324	4.265	5.826	4.051	4.648	4.553
P 049	54	Female	340	33.121	6.426	2.711	6.069	7.861	5.341	3.925	6.075	4.394	4.472	4.301
P 050	28	Female	177	30.713	4.018	6.051	6.317	6.251	6.351	3.489	7.327	4.270	4.278	4.313

4. Radiation Efficiency

Table 6:

Radiation efficiency of different antenna

Sl. No	Name of the Antenna	Frequency (GHz)	Radiation Efficiency
1	HI Antenna	2.4	18.2
2	Circular Antenna	2.0	17.8
3	HSIS Antenna	4.2	20.9
4	SAIH Antenna	4.2	21.3

SAIH antenna (21.3% efficiency) outperforms HI (18.2%), Circular (17.8%), and HSIS (20.9%) antennas for pancreas CGM. Optimized 4.2 GHz resonance, H/I slots enhance tissue coupling and E-field uniformity, while compact 38×38 mm size improves wearability over HSIS (43×43 mm).
Theoretical Calculation of SAIH Antenna

Step 1: Define Radiation Efficiency

Radiation efficiency is the ratio of radiated power to input power:

$$\eta_{rad} = \frac{P_{rad}}{P_{input}}$$

However, in practical antennas:

$$P_{input} = P_{rad} + P_{loss} + P_{reflected}$$

So, total efficiency becomes:

$$\eta_{total} = \eta_{rad}\left(1 - |S_{11}|^2\right)$$

Step 2: Calculate Reflection Coefficient from Return Loss

- Return loss = **−29.38 dB**

Convert to magnitude:

$$|S_{11}| = 10^{-29.38/20} \rightarrow |S_{11}| = 10^{-1.469} = 0.034$$

Square it: $|S_{11}| = (0.034)^2 = 0.00115$ Now: $1 + |S_{11}|^2 = 1 - 0.00115 = 0.9985$

Conclusion: Reflection loss is very small (~0.1%), so most power enters the antenna.

Step 3: Model Power Loss in Biological Tissue

In human tissue, electromagnetic waves attenuate exponentially:

$$P(z) = P_0 e^{-2\alpha Z}$$

So radiation efficiency becomes: $\eta_{\text{rad}} = e^{-2\alpha z}$

Where:

α = attenuation constant

d = penetration depth

Step 4: Calculate Attenuation Constant (α)

Formula:

$$\alpha = \sqrt{\pi f \mu \sigma}$$

Given values:

$f = 4.20GHz,\ \mu = 4\pi \times 10^{-7}$ H/m and $\sigma = 1.2S/m$

$$\alpha = \sqrt{3.14 \times 4.2 \times 10^9 \times 4 \times 3.14 \times 10^{-7} \times 1.2} = \sqrt{19900} = 141 \text{ NP/m}$$

$\alpha = 141$ NP/m

In multilayer biological tissue, effective attenuation is lower:

$\alpha_{eff} = 20 - 25\text{Np/m}$

Use Penetration Depth

Penetration depth = 3 to 4.2 cm

Take average: 3.5cm

d=3.5cm=0.035m

Step 5: Calculate Radiation Efficiency

$\eta_{\text{rad}} = e^{-2\alpha z}$

In multilayer biological tissue, effective attenuation is lower: 22

$\eta_{\text{rad}} = e^{-2\times 22\times 0.035} = e^{-1.54}$

$\eta_{\text{rad}} = 0.214 \rightarrow \eta_{\text{rad}} = 21.4\%$

Step 6: Include Matching Efficiency

$\eta_{total} = \eta_{rad}\left(1 - |S_{11}|^2\right) = 0.214 \times 0.99885 = 0.2137$

$\eta_{total} = 21.3\%$